Belfaida Mohammed Reda

+33 7 69 09 71 54 Medreda.belfaida@gmail.com
https://belfaida-portfolio.vercel.app/

Summary

Computer Science Engineer from **Grenoble INP – Ensimag**, specialized in **AI**, with industry experience in **Data Science** and **Machine Learning**. Seeking a full-time position in **AI/ML Engineering**, **Data Science**, or **AI-driven Software Engineering**.

Education

Grenoble INP - Ensimag Grenoble, France
Engineering Degree in Computer Science – MSc specialization in Artificial Intelligence (M2AI) *2023 – 2026*

- **Relevant Coursework**: Advanced statistical Learning, Machine Learning, Deep Learning, Causal Learning, Optimization, Online Learning, Natural Language Processing, Computer Vision, GenAI & Multimodal Learning

Lycée Claude Fauriel Saint-Étienne, France
Preparatory Classes, Mathematics, Physics, Computer Science *2022 – 2023*

Professional Experience

Société Générale Paris, France
Data Scientist Intern *Feb. 2026 - Aug. 2026*

- Engineered a reusable Python library to standardize credit risk modeling workflows, replacing fragmented R/SAS scripts with modular statistical components.
- Implemented statistical modeling and evaluation tools covering CART-based segmentation, model stability, bootstrap-based uncertainty estimation and risk calibration.
- Designed extensible object-oriented components and validated numerical correctness through automated testing and comparisons with reference implementations.

Wafa Gestion - Attijariwafa Bank Casablanca, Morocco
Machine Learning Intern *Jul. 2025 - Sep. 2025*

- Built an end-to-end forecasting pipeline combining time-series signals and text-derived features (news, financial publications) to predict a key macroeconomic indicator.
- Developed an NLP preprocessing pipeline for web and document data (collection, cleaning, tokenization, feature engineering) tailored to a real-world, noisy corpus.
- Implemented and compared Machine Learning forecasting models to obtain probabilistic predictions with calibrated uncertainty estimates.

Relevant Projects

Molecule Retrieval using Natural Language Queries
Natural Language Processing, Information Retrieval, Graph Learning, Contrastive Learning *Aug. 2026 - Now*

- Designed a multimodal retrieval system that aligns natural language descriptions with graph-structured objects (e.g. molecular graphs) in a shared embedding space.
- Implemented text–graph representation learning using pre-trained encoders (BERT, DistilBERT) and GNNs (GCN, GAT) with contrastive learning objectives.
- Evaluated the model using Label Ranking Average Precision (LRAP) and achieved a validation LRAP score of 96 with GCN + DistilBert architecture.

Graph-Cut Stereo Disparity Estimation and 3D Reconstruction
Computer Vision, Optimization, 3D Reconstruction *May 2026*

- Formulated stereo depth estimation as an energy minimization problem and solved it using graph cuts (Boykov–Kolmogorov max-flow/min-cut).
- Designed a 3D MRF over pixel-disparity space combining data fidelity (ZNCC-based matching cost) and smoothness constraints for globally consistent depth inference.
- Reconstructed dense depth maps and generated 3D surface meshes from rectified stereo pairs, enabling interactive visualization of scene geometry.
- Studied the effect of regularization strength on depth quality, highlighting the trade-off between smooth surfaces and sharp discontinuity preservation.

Skills

- **ML & Data:** Python, R, Scikit-learn, PyTorch, TensorFlow/Keras, Pandas, NumPy, Matplotlib, Seaborn
- **MLOps & Cloud:** Git, Github, AWS, Docker, Kafka
- **Software & Web:** C/C++, Java, JavaScript, React.js, Next.js